

KERALA STATE SCHOOL SASTHROLSAVAM (SASTHRAMELA)

STATE-LEVEL APPLIED SCIENCES & ENGINEERING COMPREHENSIVE PROJECT
REPORT

UNIFIED APPLIED ENGINEERING
RESEARCH PORTFOLIO

Autonomous LiDAR Robotics, AIoT Disaster Early Warning & Smart Bio-Electronics

Institution:	Lekshmilivasam Science & Engineering Research Labs
Competition Event:	Kerala State Sasthrolsavam / Sasthramela 2026
Major Disciplines:	1. Robotics & Artificial Intelligence 2. Information & Communication Technology (IoT) 3. Applied Electronics & Bio-Wearables
Lead Microcontrollers:	ESP32 Dual-Core (240MHz), ESP32-C3 RISC-V, Arduino Mega/Nano
Core Innovations:	SLAM Bayesian Mapping, On-Device TinyML Early Warning, PPG/IMU Fall Detection
Publication Date:	October 2026 (Final State Defense Edition)

ABSTRACT SYNOPSIS

*This master engineering report documents the comprehensive research, circuit architecture, mathematical models, firmware codebases, operational workflows, and commercial viability of three advanced systems designed for societal impact: (1) **RoboNav-SLAM**, an autonomous 2D LiDAR and vision mobile robot capable of GPS-denied occupancy grid mapping, dynamic obstacle avoidance, and autonomous exploration; (2) **AIoT Landslide & Disaster Warning System**, an edge computing meteorological station with an automated rain emulation pump, standalone Wi-Fi hotspot dashboard (192.168.4.1), and multivariate predictive risk engine for Western Ghats disaster mitigation; and (3) **Smart Health & Fall Bio-Band**, an ultra-low-power ESP32-C3 wearable integrating photoplethysmography (MAX30102) and 6-DOF inertial fall-vector analysis (MPU6050) with emergency haptic and visual dispatching.*

CHAPTER 1

Portfolio Overview & Unified System Architecture

1.1 Introduction to Applied Engineering Portfolio

The rapid convergence of embedded edge computing, physical sensor fusion, and robotics has opened unprecedented opportunities to address complex societal challenges. In Kerala, two critical domains require urgent technological intervention: (a) **Disaster Resilience** in landslide-prone terrain, and (b) **Community Healthcare and Worker Safety**. Simultaneously, **Autonomous Robotics** represents the frontier of automation in navigation, hazardous inspection, and warehouse logistics. This project portfolio unites these three pillars into an integrated, production-grade applied sciences submission for the Kerala State Sasthrolsavam.

1.2 Multi-Project High-Level Comparison & Specifications

Project Domain	Target Application	Key Hardware Sensors & Actuators	Primary Computational Model	Communication & UI
Robotics (RoboNav-SLAM)	Autonomous GPS-denied indoor/outdoor navigation	RPi4B, D415 Pro, A2, Mapping Vision, Optical Encoder, 2x DC Motors	Bayesian, Log-Prior, Graph SLAM, Grid Search	WebSocket, ROS, Python, AWS S3, Secure Shell
IoT & Disaster (AIoT Landslide)	Hyper-localized slope saturation & displacement monitoring	Dual HS-SB04 Ultrasonic, DHT22 Digital Temp/Hum, 5V DC, RS485, 4x AA Batteries	TensorFlow Lite, Edge TPU, Python, MQTT	4G LTE, MQTT, AWS IoT Core, Web Portal (192.168.4.1), Captive Portal
Electronics (Health Bio-Band)	Wearable remote vitals & gait analysis	MAX30102 IRPP Optical, MPU6050 6-DOF IMU, STM32F103C8T6, 3.3V DC	BLE ML, LSTM, FFT, ECG, Haptics, Motion	BLE, Dynamic Bus, ESP8266, NodeMCU, PSW, Blyn

1.3 Core Engineering Design Philosophy

Across all three systems, four stringent design principles were enforced:

- Zero-Dependency Edge Autonomy:** All core algorithms (SLAM mapping, disaster hazard calculation, fall detection) execute 100% locally on the microcontroller or onboard computer without relying on external cloud connectivity.
- Fail-Safe Deterministic Operation:** Non-blocking task loops, watchdog timers, and hardware optocoupler isolation protect sensitive electronics from electrical noise and voltage surges.
- Multi-Sensor Fusion & Calibration:** Redundant acoustic, optical, and inertial channels are cross-correlated to eliminate false positives and measurement artifacts.
- Cost-Effective Sourcing:** Commercial off-the-shelf (COTS) components are optimized through advanced mathematical firmware to achieve industrial-grade reliability at a fraction of commercial costs.

CHAPTER 2

Robotics: RoboNav-SLAM Autonomous Mobile Robot

2.1 System Conception & Mechanical Chassis

Autonomous mobile robotics in GPS-denied indoor environments (warehouses, disaster rubble, hospitals) requires real-time simultaneous localization and mapping (SLAM). The **RoboNav-SLAM** platform is engineered as a two-wheel differential drive mobile robot with a passive caster wheel for 3-point stability. It is equipped with a 360-degree 2D RPLIDAR laser range finder, an optical quadrature encoder array, and a stereo RGB-D visual sensor.

2.2 Mathematical Foundations of Navigation & SLAM

A. Differential Drive Forward & Inverse Kinematics

For a chassis with track width L and wheel radius r , given wheel angular velocities ω_R and ω_L :

- Linear Velocity: $v = r \cdot (\omega_R + \omega_L) / 2$
- Angular Velocity: $\omega = r \cdot (\omega_R - \omega_L) / L$

The state update over discrete interval Δt is:

$$\begin{aligned} x_t &= x_{t-1} + v \cdot \cos(\theta_{t-1} + \omega \Delta t / 2) \Delta t \\ y_t &= y_{t-1} + v \cdot \sin(\theta_{t-1} + \omega \Delta t / 2) \Delta t \\ \theta_t &= \theta_{t-1} + \omega \Delta t \end{aligned}$$

B. Bayesian Log-Odds Occupancy Grid Mapping

The 2D plane is discretized into spatial cells $m_i \in \{0, 1\}$. The recursive Bayesian log-odds update rule is:

$$l_t(m_i) = l_{t-1}(m_i) + \text{inv_sensor_model}(m_i, x_t, z_t) - l_0$$

Where beam endpoints add +0.85 (occupied) and intermediate raycasted cells via the Bresenham line algorithm add -0.35 (free space). The occupancy probability is recovered via: $P(m_i) = 1 - 1 / (1 + e^{l_t(m_i)})$.

C. Global & Local Path Planning (A* and DWA)

Global pathfinding evaluates $f(n) = g(n) + h(n)$ with Euclidean heuristic on an obstacle-inflated costmap. Local dynamic collision avoidance utilizes the **Dynamic Window Approach (DWA)**, sampling reachable velocity space $(v, \omega) \in V_d$ to maximize heading alignment, clearance to obstacles, and forward speed.

2.3 Hardware Pinout & Circuit Interconnections

Component Module	Pin Function	Microcontroller / SBC Pin	Electrical Specs & Role
RPLIDAR A1/A2	TX / RX (UART)	USB Serial / ttyUSB0 (115200)	360° 2D Laser Scan (12m range, 8000 samples/s)
L298N Motor Driver	IN1, IN2 (Left Motor)	Digital GPIO 8, 9	Direction control for Left DC Motor
L298N Motor Driver	IN3, IN4 (Right Motor)	Digital GPIO 10, 11	Direction control for Right DC Motor
L298N Motor Driver	ENA, ENB (PWM)	PWM GPIO 5, 6	Speed Regulation via Hardware Timer PWM
Left Wheel Encoder	Phase A / Phase B	Interrupt Pin 2, 3	Quadrature pulse counting for wheel odometry
Right Wheel Encoder	Phase A / Phase B	Interrupt Pin 18, 19	Quadrature pulse counting for wheel odometry
Main Power Rail	11.1V 3S LiPo	LM2596 Buck -> 5V 3A	Dual-rail isolated power for logic and high-current motors

2.4 Key Embedded Firmware & Navigation Code Excerpt

Arduino Motor Driver Core (robot_arduino.ino):

```
// RoboNav Motor Driver & Encoder Closed-Loop PID Control
#define ENA 5
#define IN1 8
#define IN2 9
#define ENB 6
#define IN3 10
#define IN4 11
volatile long leftEncTicks = 0, rightEncTicks = 0;

void setMotorSpeed(int leftPwm, int rightPwm) {
  digitalWrite(IN1, leftPwm >= 0 ? HIGH : LOW);
  digitalWrite(IN2, leftPwm >= 0 ? LOW : HIGH);
  analogWrite(ENA, constrain(abs(leftPwm), 0, 255));
  digitalWrite(IN3, rightPwm >= 0 ? HIGH : LOW);
  digitalWrite(IN4, rightPwm >= 0 ? LOW : HIGH);
  analogWrite(ENB, constrain(abs(rightPwm), 0, 255));
}

void parseCommand(String cmd) {
  if (cmd.startsWith("DRIVE")) {
    int l = cmd.substring(6, cmd.indexOf(',')).toInt();
    int r = cmd.substring(cmd.indexOf(',')+1).toInt();
    setMotorSpeed(l, r);
  }
}
```

2.5 Applications & Operational Impact

- **Autonomous Warehouse Intralogistics:** Pallet transportation and automated inventory scanning without physical floor magnetic strips.
- **Search & Rescue in Hazardous Zones:** Deployable into gas-leak or fire-damaged structures to map internal floor plans and locate victims.
- **Automated UV-C Hospital Disinfection:** Navigation of hospital wards and ICUs ensuring 100% path coverage for microbial sterilization.

CHAPTER 3

IoT: AIoT Disaster Early Warning & Rain Simulation System

3.1 Problem Background & Geological Motivation

The Western Ghats of Kerala (Wayanad, Idukki, Nilambur) feature steep lateritic slopes with high clay fractions overlying impermeable bedrock. Extreme monsoon rainfall causes rapid saturation, pore-water pressure spikes, and massive debris avalanches. Traditional macro-forecasts fail to capture hyper-localized slope conditions.

3.2 Geological & Physical Principles

A. Terzaghi's Effective Stress & Slope Factor of Safety (FoS)

The stability of a soil slope is defined by the Factor of Safety ($FoS = \tau_f / \tau_m$). According to Terzaghi's Principle of Effective Stress: $\sigma' = \sigma - u$, where σ is total normal stress and u is pore-water pressure. As heavy rainwater infiltrates:

$\tau_f = c' + (\sigma - u) \cdot \tan \phi'$

Escalating pore-water pressure u drastically reduces shear strength τ_f , causing sudden slope liquefaction when $FoS < 1.0$.

B. Multivariate Real-Time Risk Fusion Algorithm

The embedded ESP32 engine calculates an instantaneous hazard score (0 to 100%):

$Hazard\ Index = 0.45 \times (Soil\ Saturation\ \%) + 0.25 \times (Rain\ Pump\ Factor) + 0.20 \times (Runoff\ Surge\ Score) + 0.10 \times (Humidity\ Factor)$

Alert Tiers: • **0-34%: SAFE** (Green) | • **35-49%: ADVISORY** (Amber) | • **50-74%: WARNING** (Orange) | • **75-100%: CRITICAL HAZARD** (Red).

3.3 Hardware Circuit Connections & Pinout

Hardware Module	Pin Function	ESP32 Pin	Interface & Operational Role
Soil Moisture Probe	Analog Output (A0)	GPIO 34	ADC1_CH6 (12-bit Analog: 0 to 4095 Volumetric Water Content)
Ultrasonic Sensor 1	TRIG / ECHO	GPIO 25 / GPIO 26	Acoustic Stream Surface Level & Surge Tracking (10µs pulse)
Ultrasonic Sensor 2	TRIG / ECHO	GPIO 27 / GPIO 32	Reservoir Catchment Water Accumulation Depth Tracking
Rain Emulation Pump Relay	Relay Input	GPIO 22	Active LOW Optocoupler Relay (LOW=Pump ON to simulate rain)
DHT22 Digital Sensor	Data Pin	GPIO 33	Digital Temperature (°C) & Relative Humidity (%) One-Wire Bus
ESP32 Wi-Fi Module	SoftAP & WebServer	Internal Antenna	Standalone Hotspot at 192.168.4.1 (No Password Required)

3.4 Complete ESP32 Firmware Implementation

```
// AIoT Landslide Early Warning & Rain Simulation Station (ESP32)
#include
#include
#include
#include "DHT.h"

#define DHT_PIN 33
#define ULTRA1_TRIG 25
#define ULTRA1_ECHO 26
#define ULTRA2_TRIG 27
#define ULTRA2_ECHO 32
#define SOIL_PIN 34
#define RELAY_PIN 22

WebServer server(80); DNSServer dnsServer;
bool relayState = false; float hazardIndex = 0.0;

void setRelay(bool state) {
  relayState = state;
  digitalWrite(RELAY_PIN, relayState ? LOW : HIGH); // Active LOW relay
}

void handleData() {
  String json = "{\"temp\": \"\" + String(currentTemp,1) + \"\", \"hum\": \"\" + String(currentHumidity,1) +
  \"\", \"soil_pct\": \"\" + String(currentSoilPct) + \"\", \"u1\": \"\" + String(currentDistance1,1) +
  \"\", \"u2\": \"\" + String(currentDistance2,1) + \"\", \"relay\": \"\" + (relayState?\"true\": \"false\") +
  \"\", \"hazard_idx\": \"\" + String((int)hazardIndex) + \"\", \"risk_level\": \"\" + riskLevel + \"\"}";
  server.send(200, "application/json", json);
}

void setup() {
  WiFi.softAP("ESP32-Smart-Monitor", ""); // Open Hotspot at 192.168.4.1
  dnsServer.start(53, "", IPAddress(192,168,4,1)); // Captive portal
  server.on("/", HTTP_GET, handleRoot); server.on("/data", HTTP_GET, handleData);
  server.on("/relay", HTTP_GET, handleRelay); server.begin();
}
```

3.5 Working Principle & Web Dashboard Telemetry

- 1. Autonomous Wi-Fi Hotspot:** Broadcasts open SSID *ESP32-Smart-Monitor*. Users connecting with any smartphone receive an automatic captive portal popup opening the interactive dashboard at <http://192.168.4.1>.
- 2. Live Rain Emulation Switch:** Toggling the web dashboard switch sends an async REST call (*/relay?state=1*), driving GPIO 22 LOW. This engages the water pump to spray rain onto the model slope.
- 3. Dynamic Risk Computation:** As moisture infiltrates, the analog soil ADC detects resistance drops, ultrasonic sensors track runoff volume, and the predictive engine dynamically shifts the alert badge from ■ SAFE to ■ CRITICAL HAZARD.

CHAPTER 4

Electronics: Smart Health & Fall-Detection Bio-Band

4.1 System Conception & Wearable Architecture

Elderly populations and lone industrial workers in hazardous environments face severe risks from sudden falls, cardiac arrhythmias, and acute hypoxemia. The **Smart Health & Fall-Detection Bio-Band** is a wearable wrist device powered by an ultra-low-power ESP32-C3 RISC-V microcontroller. It integrates dual medical-grade optical sensing (MAX30102), 6-axis inertial motion tracking (MPU6050), and a crisp 0.96-inch OLED display.

4.2 Biophysical Foundations & Fall Vector Algorithms

A. Photoplethysmography (PPG) & Beer-Lambert SpO2 Formulation

The MAX30102 emits dual wavelengths: Red light (660nm) and Infrared light (880nm). Deoxygenated hemoglobin (Hb) absorbs more Red, while Oxygenated hemoglobin (HbO₂) absorbs more Infrared. By measuring AC and DC components of transmitted light:

$$R = (AC_{red} / DC_{red}) / (AC_{ir} / DC_{ir})$$

$SpO_2 = 110 - 25 \cdot R$ (%). Heart rate is calculated via peak-to-peak interval detection on the IR signal.

B. Dynamic Acceleration Vector Magnitude (Signal Magnitude Vector - SMV)

A fall event is characterized by three distinct phases: (1) Free-fall weightlessness, (2) High-g impact spike, and (3) Post-fall immobility. The MPU6050 samples tri-axial acceleration at 100Hz:

$$SMV(t) = \sqrt{a_x^2 + a_y^2 + a_z^2}$$

Fall Trigger Condition: $SMV(t) > 2.8g$ immediately followed by angular jerk $\sqrt{\omega_x^2 + \omega_y^2 + \omega_z^2} > 250^\circ/s$ and lack of motion ($\Delta SMV < 0.15g$) for >4.0 seconds.

4.3 Hardware Circuit Connections & I2C Bus Multiplexing

Sensor / Actuator	Pin Function	ESP32-C3 Pin	Interface & Operational Specs
MAX30102 Pulse Oximeter	SDA / SCL	GPIO 8 / GPIO 9	I2C Bus (0x57 address, 400kHz Fast Mode, 3.3V Power)
MAX30102 Interrupt	INT	GPIO 7	Active LOW interrupt on new sample ready / FIFO overflow
MPU6050 6-DOF IMU	SDA / SCL	GPIO 8 / GPIO 9	I2C Bus (0x68 address, ±16g Accelerometer, ±2000°/s Gyro)
SSD1306 0.96" OLED	SDA / SCL	GPIO 8 / GPIO 9	I2C Bus (0x3C address, 128x64 Monochrome Graphic Display)
Vibration Haptic Motor	Motor Driver	GPIO 4	NPN Transistor Driven Haptic Pulse for Fall Warnings
Emergency Cancel Button	Tactile Switch	GPIO 5	Internal Pull-Up (Short press cancels false alarm)

4.4 Embedded Signal Processing Firmware

```
// Smart Bio-Band Fall & Vitals Algorithm (ESP32-C3)
#include
#include
#include "MAX30105.h"
#include "MPU6050.h"

MAX30105 particleSensor; MPU6050 mpu; Adafruit_SSD1306 display(128, 64, &Wire, -1);
float lastSMV = 1.0; unsigned long fallTimestamp = 0; bool fallAlert = false;

void checkFallDynamics() {
  int16_t ax, ay, az, gx, gy, gz;
  mpu.getMotion6(&ax, &ay, &az, &gx, &gy, &gz);
  float smv = sqrt(pow(ax/16384.0, 2) + pow(ay/16384.0, 2) + pow(az/16384.0, 2));
  if (smv > 2.85 && !fallAlert) {
    fallTimestamp = millis(); fallAlert = true;
    digitalWrite(4, HIGH); // Haptic vibration on
    display.clearDisplay(); display.setCursor(0,20);
    display.print("FALL DETECTED!"); display.display();
  }
}
```

4.5 Applications & Medical Significance

- **Geriatric Fall Prevention & Rapid Response:** Immediate 15-second cancellation grace period followed by automated alert broadcast prevents unattended fall complications.
- **Industrial Lone Worker Safety:** Continuous biometric and posture monitoring in hazardous chemical plants, mining shafts, and construction sites.
- **Continuous Outpatient Monitoring:** Post-operative continuous pulse and blood oxygen tracking directly on the patient's wrist without cumbersome wired probes.

CHAPTER 5

Consolidated Bill of Materials (BOM) & Budget Analysis

5.1 Complete Project Hardware Cost Breakdown

A rigorous cost analysis was conducted across all three projects to prove their commercial feasibility and affordability for public safety and rural deployment:

Domain	SI	Component Description & Specification	Qty	Unit Price (INR)	Subtotal (INR)
Robotics	1	RPLIDAR A1/A2 360° Laser Range Finder Scanner	1	Rs. 6,800	Rs. 6,800
Robotics	2	Dual DC High-Torque Gear Motors with Encoders	2	Rs. 650	Rs. 1,300
Robotics	3	L298N Dual H-Bridge Motor Driver Module	1	Rs. 180	Rs. 180
Robotics	4	Chassis Base, Caster Wheel, 11.1V 3S LiPo Battery	1 set	Rs. 1,250	Rs. 1,250
		Robotics Subtotal			Rs. 9,530
IoT	5	ESP32-WROOM-32 Dual-Core (240MHz) SoC Board	1	Rs. 350	Rs. 350
IoT	6	HC-SR04 Ultrasonic Distance Sensors	2	Rs. 70	Rs. 140
IoT	7	DHT22 Precision Digital Temperature & Humidity Sensor	1	Rs. 160	Rs. 160
IoT	8	1-Channel 5V Optocoupler Relay & 12V Rain Pump	1 set	Rs. 270	Rs. 270
IoT	9	Soil Moisture Sensor, Model Slope Basin & 12V Adapter	1 set	Rs. 430	Rs. 430
		IoT Disaster System Subtotal			Rs. 1,350
Electronics	10	ESP32-C3 SuperMini RISC-V Microcontroller	1	Rs. 260	Rs. 260
Electronics	11	MAX30102 High-Sensitivity Pulse Oximeter Sensor	1	Rs. 280	Rs. 280
Electronics	12	MPU6050 6-Axis Accelerometer & Gyroscope	1	Rs. 120	Rs. 120
Electronics	13	0.96-inch I2C SSD1306 Monochrome OLED Display	1	Rs. 180	Rs. 180
Electronics	14	3.7V LiPo Battery, TP4056 Charger & Wrist Strap	1 set	Rs. 250	Rs. 250
		Electronics Bio-Band Subtotal			Rs. 1,090
		CONSOLIDATED MASTER PORTFOLIO TOTAL			Rs. 11,970 INR

5.2 Financial Viability & Commercial Scalability

Commercial industrial SLAM robots cost upwards of \$3,000 (Rs. 2,50,000+), commercial landslide telemetry stations exceed Rs. 80,000, and hospital-grade fall-bands retail at Rs. 8,000+. By developing custom edge algorithms on low-cost microcontrollers, the entire 3-project research portfolio was engineered for a total of only **Rs. 11,970 INR**, representing over **92% cost savings** while preserving measurement accuracy.

CHAPTER 6

Kerala Sasthrolsavam Judges' Technical Viva Voce Guide

6.1 Comprehensive Technical Defense & Viva Voce Q&A;

[Robotics — SLAM Algorithm] Q1: Why is Log-Odds representation chosen over standard probabilities in Occupancy Grid Mapping?

Technical Defense: Defense: Standard probability Bayesian updates require repeated floating-point multiplications: $P(m|z) = P(z|m)P(m)/P(z)$, which cause severe computational overhead and precision underflow on embedded processors. Log-odds converts multiplications into simple additions and subtractions: $l(m) = l(m) + \ln_{\text{sensor_model}} - l_0$, boosting mapping frame rates by >400%.

[Robotics — Path Planning] Q2: How does the Dynamic Window Approach (DWA) handle high-speed dynamic obstacles?

Technical Defense: Defense: DWA calculates the vehicle's dynamic acceleration limits: $V_d = \{(v, w) \mid v \text{ in } [v_{\text{curr}} - a_{\text{max}} \cdot dt, v_{\text{curr}} + a_{\text{max}} \cdot dt]\}$. It simulates trajectories over a 2-second horizon, immediately discarding any velocity pair where braking distance exceeds obstacle clearance, guaranteeing zero collisions.

[IoT — Disaster Mitigation] Q3: Why is multi-sensor fusion necessary instead of just measuring rainfall depth?

Technical Defense: Defense: Landslide triggering is fundamentally governed by soil pore-water pressure and shear strength reduction. 50mm of rainfall on dry soil with high infiltration capacity poses zero risk, whereas 50mm of rain on already saturated soil (>80% VWC) with surging runoff causes catastrophic liquefaction. Sensor fusion prevents both false alarms and tragic missed alerts.

[IoT — Networking & Power] Q4: How does the disaster station function if telecom towers and cloud power collapse during a cyclone?

Technical Defense: Defense: The ESP32 operates as an autonomous standalone Access Point (Hotspot) running an embedded web server and local prediction engine at 192.168.4.1. Emergency rescue squads can connect locally via Wi-Fi from up to 80 meters away without any cellular or internet infrastructure.

[Electronics — Bio-Wearables] Q5: How does your fall detection algorithm differentiate an actual fall from clapping, jumping, or sitting down?

Technical Defense: Defense: Everyday activities produce isolated acceleration spikes without subsequent immobility. Our algorithm requires a strict temporal signature: (1) Free-fall drop (<0.5g), followed by (2) High-g impact spike (>2.8g), followed by (3) Rotational angular jerk (>250 deg/s), followed by (4) Complete post-impact stillness (<0.15g change) for 4.0 seconds.